ProgRec: Undergraduate Research Program Mentor Recommend Agent

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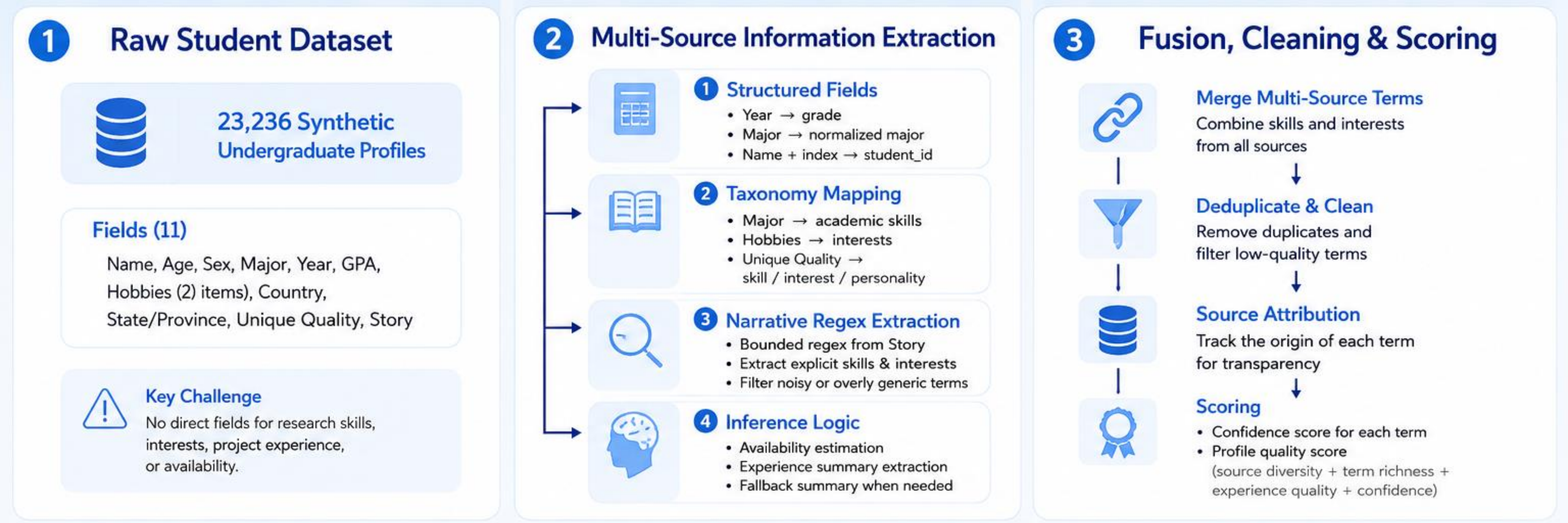
Agent Introduction

Skill 1: Student Profiling Skill

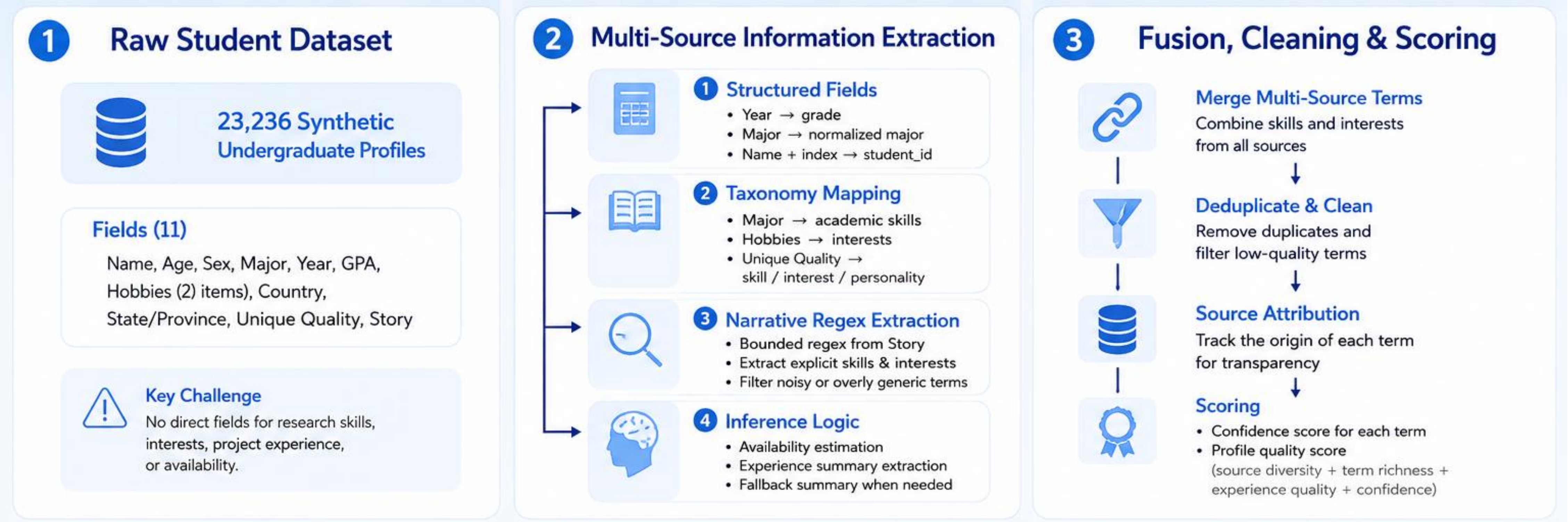


Input: Raw Student Dataset

Output:

Function:

Skill 1: Student Profiling Skill



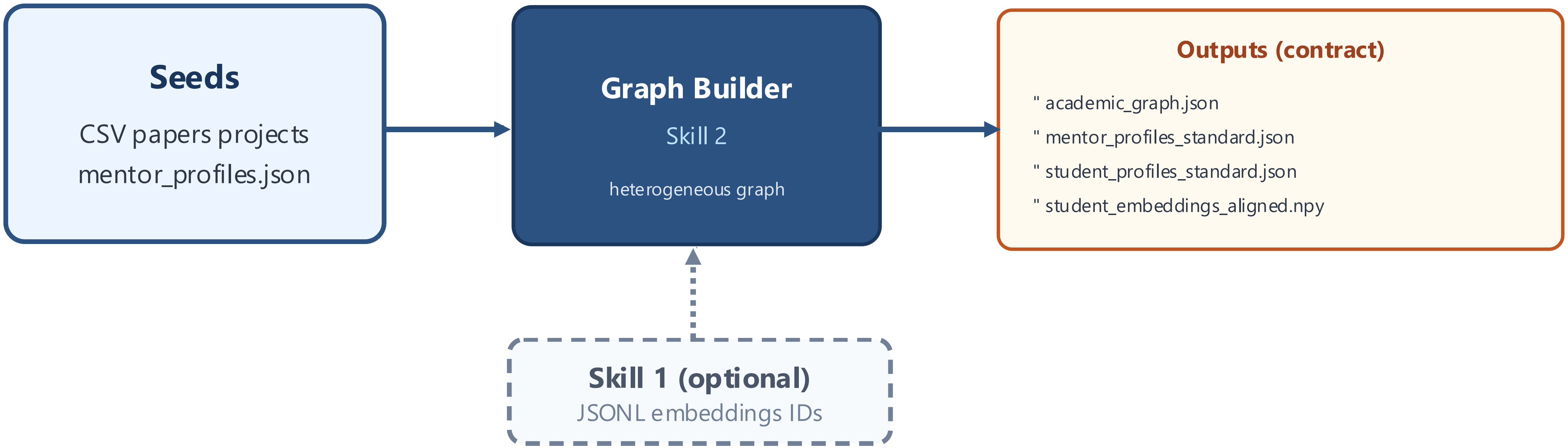
Input: Raw Student Dataset

Output:

Function:

Skill 2: Academic Graph Builder Skill Pipeline

Inputs: seeded profiles + papers/projects **(+ optional Skill 1 layer)**
Outputs: contract files for later skills—**no need to import Skill 2 code**



Heterogeneous Graph: Nodes, Edges and Construction

Nodes (5 types)

Mentor
Paper
Topic
Project
Student

One JSON bucket per type in academic_graph.json

Edges (grouped)

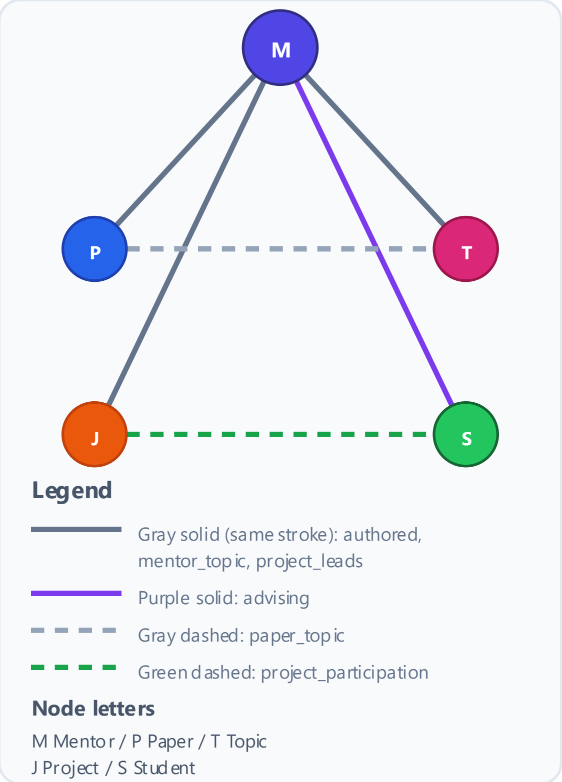
Academic
authored, paper_topic, mentor_topic, topic_similarity, collaboration

Campus
project_leads, project_participation, advising

Peer
skill_complementarity, shared_interest

Each edge carries type + weight for downstream SNA / explanations

Schematic



Mentor–topic links are aggregated via paper chains; student–student edges are capped at scale. In alignment with Skill 1 mode, advising/participation are resynthesized for consistent student IDs.

Skill 2: Deliverables and Skill 1 Alignment

Deliverable	Purpose
academic_graph.json	**One graph** for analysis & explanations
mentor_profiles_standard.json	**Mentor profiles** ready for retrieval / embeddings
student_profiles_standard.json	**Student profiles** aligned with Skill 1
embeddings + ids	**Vectors line up with** the exported student list

Skill 1 adds real students + embeddings; otherwise Skill 2 runs on seeded CSVs.
Aligned vectors are a subset indexed from Skill 1—no retraining.

Skill 3: Mentor Discovery Skill

Function:

Input:

Output:

Skill 4: Project & Teammate Discovery Skill

Function: Project and Teammate Recommendation

$$
\begin{aligned}
\text{Teammate Score} = {} & 0.35 \times \text{Shared Interest} \\
& + 0.35 \times \text{Skill Complementarity} \\
& + 0.10 \times \text{Availability} \\
& + 0.20 \times \text{Graph Relation}
\end{aligned}
$$

Input:Target student profile, Skill 2 academic graph, Skill 3 mentor candidates, Student candidate pool

Output: Related project, Recommended teammates

Skill 5: Social Ranking Skill

Function:

Mentor Candidates

Project Resources

Potential Teammates

Scoring

Final Recommendation

Key Method:

- Maximal Marginal Relevance (MMR) diversity re-ranking

$$MMR = \operatorname{Arg} \max_{d_i \in R \setminus S} \left[\lambda \cdot \mathrm{Sim}_1(d_i, Q) - (1 - \lambda) \cdot \max_{d_j \in S} \mathrm{Sim}_2(d_i, d_j) \right]$$

Skill 5: Social Ranking Skill

Highlight:

- Explainable Output: Reason
- Robustness: Cold Start: + cold start bonus

Mentor	Publication: empty	Topic < 2	
Student	Skills < 2	Experience_summary < 20 words	Interests < 2

Summary